

# MedTrack\_DV (Hospital Operations & Patient Analytics Dashboard)

**Project Statement:** This project aims to develop a comprehensive hospital operations and patient analytics dashboard suite to analyse hospital performance, patient admissions, department efficiency, and healthcare resource utilisation. Using publicly available Hospital Datasets and Patient Admission Datasets, the project will transform healthcare operational data into actionable insights through interactive Tableau dashboards. The dashboard suite will help hospital administrators, healthcare managers, medical staff, and policymakers monitor patient flow, evaluate department performance, optimise hospital resources, and improve operational efficiency through data-driven decision-making.

## Dataset Resources:

### Dataset Name: Hospital HMIS Dataset for Healthcare Analytics

**Dataset Description:** This dataset represents a synthetic but realistic Hospital Management Information System (HMIS) designed for healthcare analytics, dashboards and operational insights. It contains multiple interconnected tables covering patient information, admissions, departments, wards, beds, billing, insurance and staff management.

The dataset is designed to represent the functioning of a hospital management information system. The **admission table acts as the central transactional table**, which can be connected with patient, department, ward, disease and billing-related tables using common keys.

## Tables Available in the Original Dataset

The original Kaggle dataset contains the following tables:

1. department.csv
2. ward.csv
3. bed.csv
4. disease.csv
5. diagnostic\_test.csv
6. drug\_manufacturer.csv
7. insurance\_provider.csv
8. patient.csv
9. admission.csv
10. billing.csv
11. billing\_detail.csv
12. patient\_diagnostic.csv
13. drug.csv
14. drug\_inventory.csv
15. prescription.csv

16. patient\_insurance.csv
17. employee.csv
18. doctor.csv
19. staff\_assignment.csv

**Dataset Link:** <https://www.kaggle.com/datasets/shalakagangurde/hospital-hmis-dataset-for-healthcare-analytics>

**Source:** Kaggle

### Tables Used in This Project:

For this project, the tables that were useful for hospital operations analysis were selected and integrated into a consolidated dataset.

- **Patient table** – Used for patient details and demographics.
- **Admission table** – Used for admission, discharge, patient, department, ward and disease details.
- **Department table** – Used for department names and analysis.
- **Ward table** – Used for ward details, ward types and bed capacity.
- **Bed table** – Used for bed information and resource utilisation.
- **Disease table** – Used for disease names and categories.
- **Billing table** – Used for billing, insurance and patient payable amounts.
- **Employee table** – Used for employee details and staff count.
- **Staff Assignment table** – Used for department/ward-wise staff allocation and shifts.

| Table                | Columns Used                                                                                                                           |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| patient.csv          | patient_id, gender, date_of_birth, blood_group, city, contact_number                                                                   |
| admission.csv        | admission_id, admission_date, discharge_date, admission_type, admission_status, patient_id, department_id, ward_id, bed_id, disease_id |
| department.csv       | department_id, department_name, department_type, floor_number, status                                                                  |
| ward.csv             | ward_id, ward_name, ward_type, total_beds, department_id                                                                               |
| bed.csv              | bed_id, bed_number, bed_status, ward_id                                                                                                |
| disease.csv          | disease_id, disease_name, disease_category                                                                                             |
| billing.csv          | bill_id, bill_date, total_amount, insurance_covered_amount, patient_payable_amount, payment_status, payment_mode, admission_id         |
| employee.csv         | employee_id, employee_name, gender, role, employment_type, date_of_joining, department_id                                              |
| doctor.csv           | doctor_id, employee_id, specialization, qualification, experience_years                                                                |
| staff_assignment.csv | assignment_id, employee_id, ward_id, shift                                                                                             |

### Data Integration:

The selected tables were integrated using their common key fields. The Admissions table was used as the primary transaction table, while the Patients, Departments, Wards, Beds, Disease, Billing, Employee, and Staff Assignment tables were connected through their respective identifiers.

The main relationships used during integration were:

- patient\_id connects Patients and Admissions
- department\_id connects Admissions, Departments, Wards, and Employee

- ward\_id connects Admissions, Wards, Beds, and Staff Assignment
- bed\_id connects Admissions and Beds
- disease\_id connects Admissions and Disease
- admission\_id connects Admissions and Billing
- employee\_id connects Employee and Staff Assignment

The Integrated Dataset has 45000 rows and 34 columns. This dataset has been used in further analysis, such as cleaning and transformation.